KUMAR ANURAG SAHU

Software Engineer – GenAI / Agentic AI

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SUMMARY

Software Engineer specializing in Generative AI, LLM applications, and Agentic AI systems, with hands-on experience building production AI for enterprise freight and logistics workflows. Experienced in Mistral LLM integration, Retrieval-Augmented Generation (RAG), embeddings, vector search, semantic search, reranking, prompt engineering, structured function calling, tool calling, custom agent architectures, human-in-the-loop workflows, AI guardrails, and evaluation harnesses, backed by strong full-stack engineering in TypeScript, Node.js, React, MongoDB, REST APIs, event-driven systems, and AWS.

CORE SKILLS

Generative AI & LLMs: Large Language Models (LLMs), Generative AI, Mistral, LLM Integration, Prompt Engineering, Structured Outputs, Function Calling, Tool Calling, Context-Window Management, AI Document Automation

RAG & Retrieval: Retrieval-Augmented Generation (RAG), Embeddings, Semantic Search, Vector Search, kNN Retrieval, Reranking, MongoDB Atlas Vector Search, Voyage Embeddings, Voyage Reranking, Few-Shot Retrieval, LlamaParse

Agentic AI & Reliability: AI Agents, Agentic Workflows, Custom Agent Architecture, Agent Loops, Tool Registry, Tool Dispatch, Human-in-the-Loop, Approval Workflows, Capability Gates, Action Gates, AI Guardrails, Evaluation Harnesses, Golden Datasets, Model Canaries, Confidence Scoring, Retry Logic, Audit Logging, Observability

Engineering & Cloud: TypeScript, JavaScript, Node.js, Express.js, MongoDB, Mongoose, REST APIs, Zod, Server-Sent Events (SSE), Async Processing, Event-Driven Architecture, React 18, Next.js, Vite, RTK Query, AWS S3, AWS SQS FIFO, Docker, Valkey, CI/CD, Jest

PROFESSIONAL EXPERIENCE

REUDAN INTERNATIONAL

Software Engineer – GenAI / Agentic AI | November 2024 – Present

Generative AI & Agentic AI

- Designed and built Kandyr Copilot, a production AI agent platform embedded in a freight-operations application, enabling natural-language interaction with enterprise data, document workflows, artifact generation, and gated business actions.
- Built a custom agentic AI architecture with a server-side agent loop, policy-controlled execution (interactive and headless modes, bounded step limits), tool registry, tool dispatch, structured tool results, and SSE streaming; integrated Mistral LLMs for tool calling, structured function calls, and document extraction.
- Developed a natural-language-to-MongoDB query engine in which Mistral generates a constrained intermediate representation, validated server-side against a field/operator allow-list and organization scope before compilation into MongoDB aggregation pipelines.
- Designed reusable LLM tool-calling infrastructure with typed input schemas, capability checks, side-effect classification (read, write, external, artifact), preview/commit phases, and audit logging; built tools for querying, document extraction, file generation, email, entity drafts, and readiness checks.
- Implemented human-in-the-loop workflows where write and external actions produce an approval proposal and commit only after explicit approval, with commit-time re-resolution of facts and gate re-evaluation; added retries with exponential backoff, tool-call correlation, and context-window management via result summarization and compact history digests.

RAG, Vector Search & Document AI

- Built an AI document-automation system using LlamaParse and Mistral to extract structured logistics and accounting data from PDFs, with multi-document extraction, file-type tagging, engineered prompt instructions, source-document preservation, and structured function-call schemas.
- Implemented semantic entity resolution using MongoDB Atlas Vector Search, embeddings, fuzzy-matching fallback, and Voyage reranking to match extracted vendors, charges, sites, and ports against master data; applied RAG with Voyage embeddings and kNN retrieval of similar historical questions as few-shot context.

AI Guardrails, Safety & Evaluation

- Designed an AI-assisted purchase-invoice auto-conversion workflow with confidence thresholds, bank-account validation, duplicate detection, rate/quantity validation, provenance tracking, and human approval, plus a feedback loop converting validated corrections into vendor and charge aliases for future matching.
- Implemented fail-closed guardrails, capability and action gates, tenant-isolation controls, prompt-level domain grounding and refusal rules, and AI action audit logging capturing tool, side-effect type, gate decisions, latency, outcomes, and human corrections.
- Developed approximately 20 server-side and 12 client-side test suites for Copilot and AI workflows, plus a golden evaluation corpus, RAG seed corpus, and weekly real-Mistral canary to detect LLM regressions.

Full-Stack AI Product & Event-Driven Platform

- Built the React-based Copilot interface with streaming conversations, tool-call and approval cards, entity drafts, attachments, and generated artifacts; implemented 19 Copilot API routes including 3 SSE endpoints, consumed via 14 RTK Query client endpoints.
- Designed an event-driven ETL pipeline using Valkey and AWS SQS FIFO to process booking lifecycle events into a pre-aggregated daily MongoDB analytics cube, with frozen FX rates for financial correctness and full-rebuild/dry-run reconciliation workflows.

CLEVERATTI SKILLS PVT LIMITED

Junior Full Stack Developer Intern | May 2024 – August 2024

- Analyzed an existing live codebase and independently built a complete web application, frontend and backend, using the MERN stack.

EDUCATION & CERTIFICATIONS

Bachelor of Technology (B.Tech), Computer Science – SIET, Dhenkanal | 2020 – 2024

Full Stack Web Development Course – 100xDevs | Issued September 2024 | Credential ID: 0FFKL29X